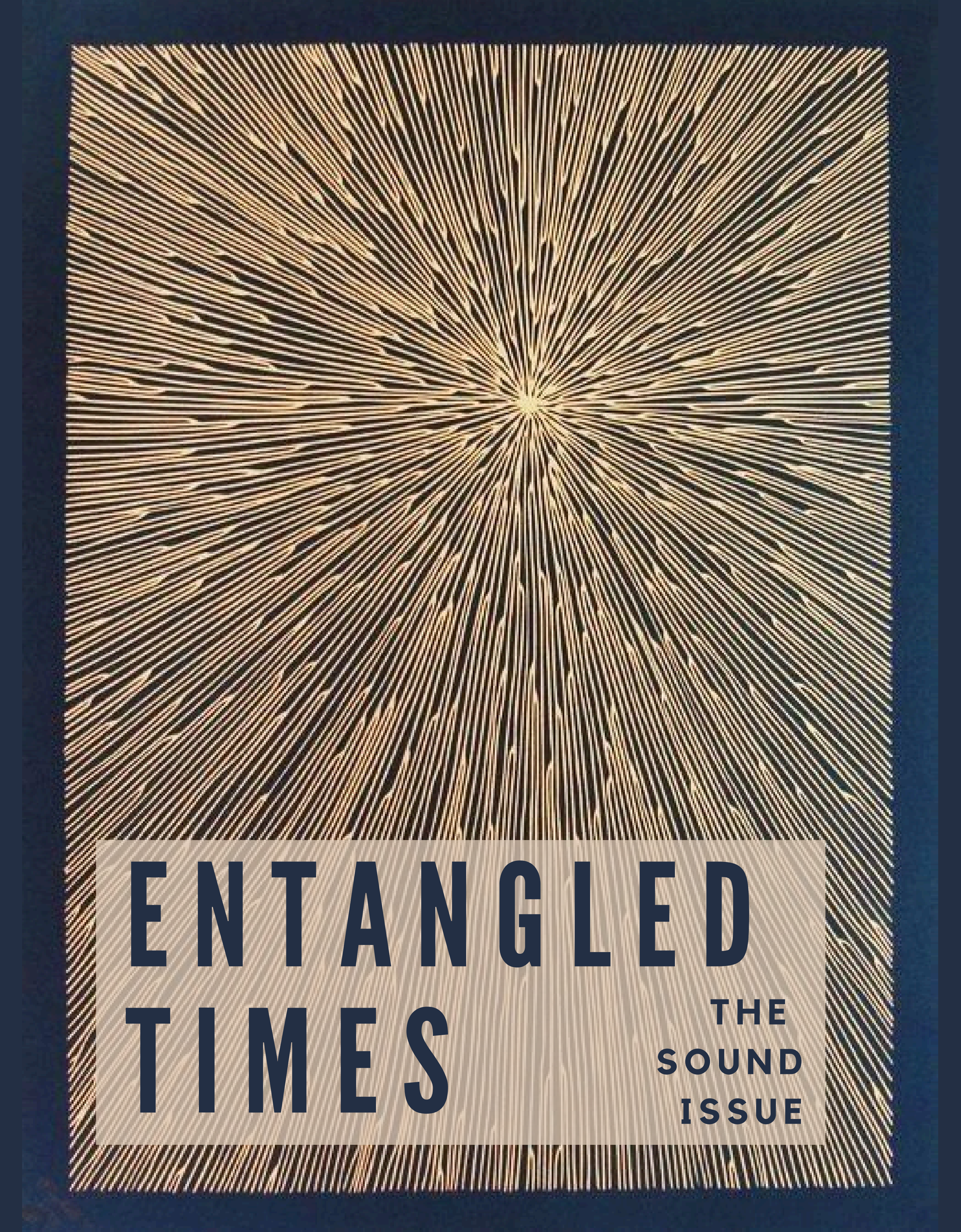




ENTANGLED TIMES

THE
SOUND
ISSUE

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WHAT IS SOUND?



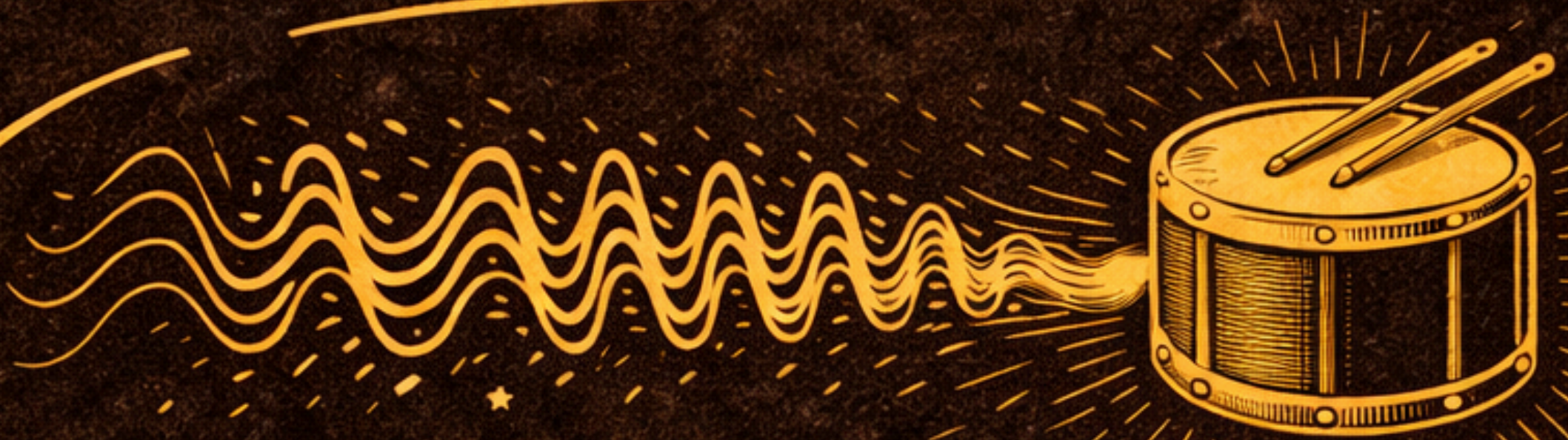
SOUND IS A TYPE OF VIBRATION THAT TRAVELS THROUGH THE AIR (OR OTHER MATERIALS) AND ALLOWS US TO HEAR. WHEN SOMETHING VIBRATES, LIKE A GUITAR STRING, A DRUM, OR YOUR VOCAL CORDS, IT CREATES SOUND.

SOUND MOVES AS WAVES, CARRYING VIBRATIONS FROM ONE PLACE TO ANOTHER. WHEN THESE VIBRATIONS REACH OUR EARS, OUR BRAIN TURNS THEM INTO SOUNDS LIKE MUSIC, VOICES, OR NOISE!

UNLIKE LIGHT, SOUND CANNOT TRAVEL THROUGH EMPTY SPACE. THAT'S WHY SPACE IS SILENT!



HOW SOUND IS MADE?



SOUND IS CREATED WHEN AN OBJECT VIBRATES BACK AND FORTH.

EXAMPLES:

- WHEN YOU HIT A DRUM, THE SURFACE VIBRATES.
- WHEN YOU SPEAK, YOUR VOCAL CORDS VIBRATE.
- WHEN YOU PLUCK A GUITAR STRING, IT VIBRATES.

THESE VIBRATIONS PUSH THE AIR AROUND THEM, CREATING SOUND WAVES THAT MOVE OUTWARD IN ALL DIRECTIONS.

HOW SOUND TRAVELS?

SOUND TRAVELS IN THE FORM OF WAVES. UNLIKE LIGHT, SOUND NEEDS A MEDIUM TO TRAVEL THROUGH. THIS MEANS SOUND CANNOT TRAVEL THROUGH EMPTY SPACE. SOUND CAN MOVE THROUGH:

- AIR (TALKING, MUSIC)
- WATER (WHALES COMMUNICATING)
- SOLIDS (TAPPING A TABLE AND HEARING IT FROM THE OTHER SIDE)

SOUND TRAVELS FASTEST IN SOLIDS, SLOWER IN LIQUIDS, AND SLOWEST IN GASES.

FUN FACT: SOUND CAN TRAVEL FASTER THROUGH SOLIDS BECAUSE THE PARTICLES ARE CLOSER TOGETHER.

PITCH

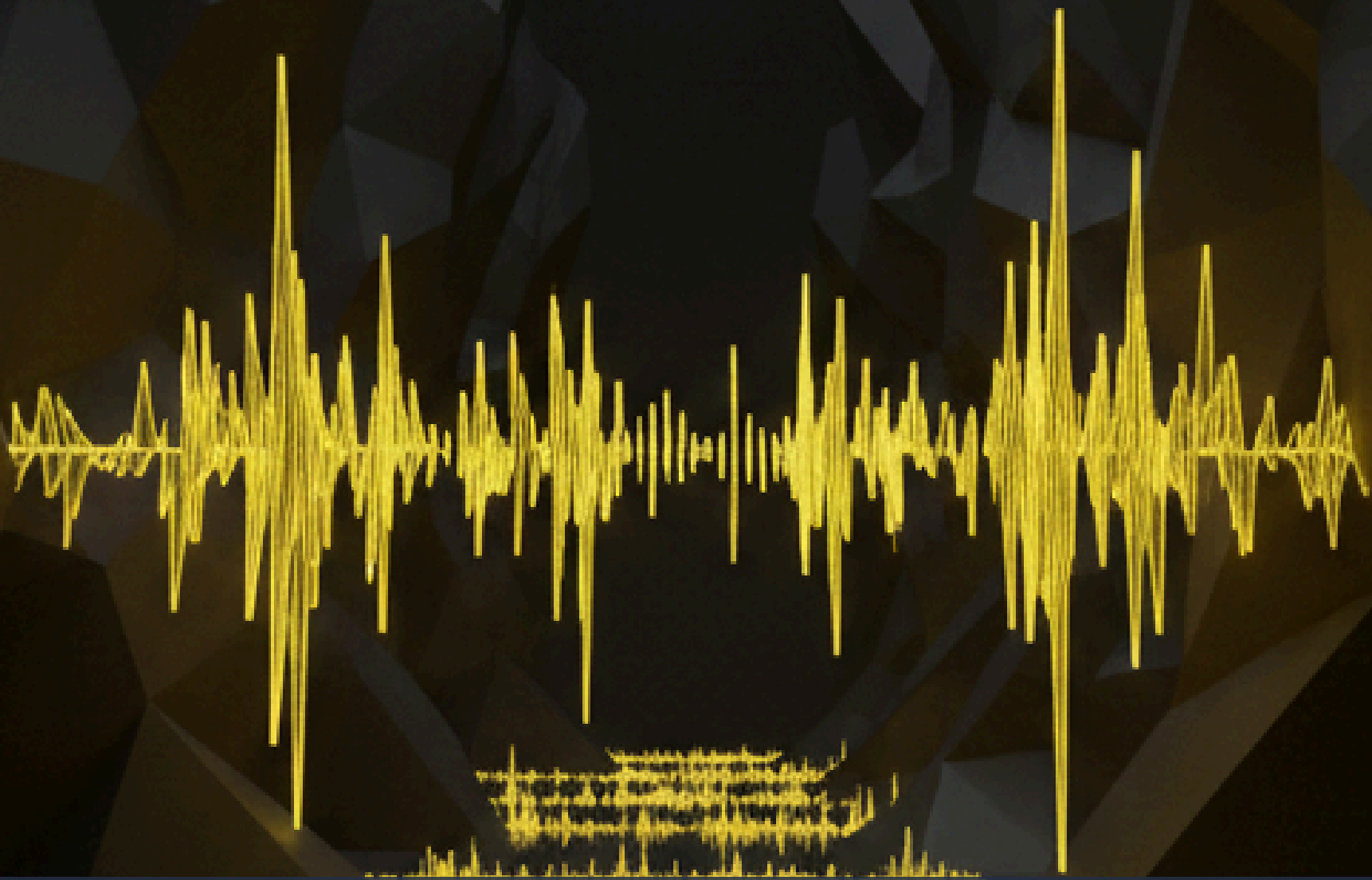
TELLS US HOW HIGH OR LOW A SOUND IS. FAST VIBRATIONS CREATE A HIGH PITCH (LIKE A WHISTLE), WHILE SLOW VIBRATIONS CREATE A LOW PITCH (LIKE A DRUM).



LOUDNESS

TELLS US HOW STRONG A SOUND IS. BIGGER VIBRATIONS PRODUCE LOUDER SOUNDS, AND SMALLER VIBRATIONS PRODUCE SOFTER SOUNDS. LOUDNESS IS MEASURED IN DECIBELS (DB).

ECHOES AND REFLECTION OF SOUND



WHEN SOUND HITS A HARD SURFACE, IT CAN BOUNCE BACK. THIS BOUNCING OF SOUND IS CALLED AN ECHO. ECHOES ARE COMMONLY HEARD IN:

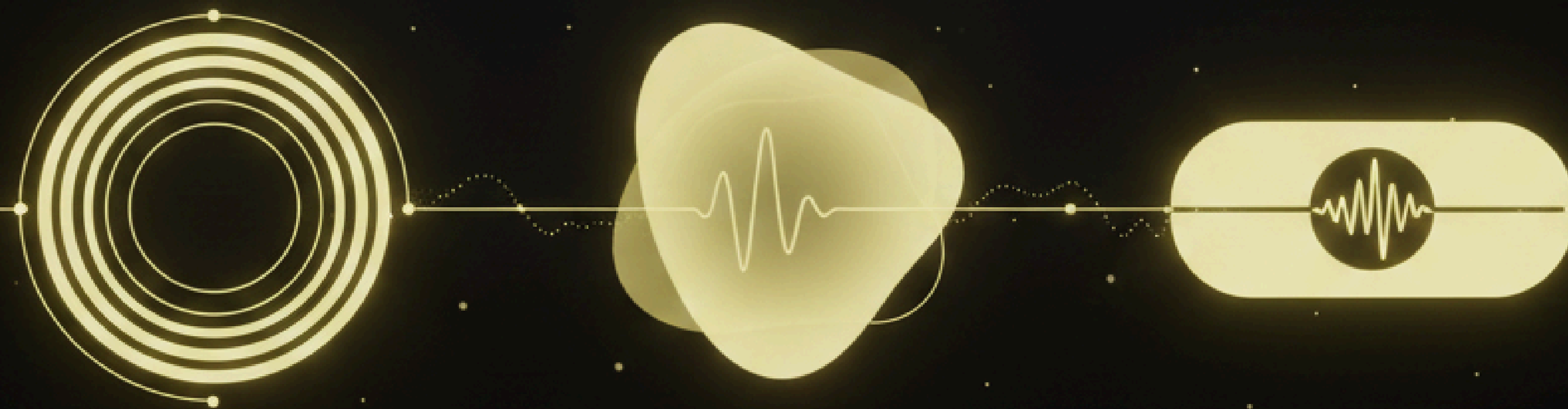
- EMPTY ROOMS
- MOUNTAINS
- TUNNELS

SOFT MATERIALS LIKE CURTAINS, CARPETS, AND CUSHIONS ABSORB SOUND, WHICH IS WHY ROOMS WITH FURNITURE DO NOT ECHO MUCH.

HOW WE USE SOUND

MEDICINE:
ULTRASOUND
SCANS HELP
DOCTORS SEE
INSIDE THE
HUMAN BODY

COMMUNICATION:
TALKING,
PHONE CALLS,
VIDEO CALLS



SAFETY:
SIRENS AND
ALARMS WARN
US OF DANGER

MUSIC:
SINGING,
INSTRUMENTS,
CONCERTS

NAVIGATION:
BATS AND
DOLPHINS USE
SOUND TO
LOCATE OBJECTS

FUN SOUND EXPERIMENTS

RUBBER BAND GUITAR

STRETCH RUBBER BANDS OVER AN EMPTY BOX AND PLUCK THEM. NOTICE HOW THICKER RUBBER BANDS PRODUCE LOWER SOUNDS.

ECHO TEST

CLAP YOUR HANDS IN AN EMPTY ROOM OR HALLWAY AND LISTEN CAREFULLY FOR AN ECHO.

FEEL THE VIBRATIONS

PLACE YOUR HAND ON YOUR THROAT WHILE SPEAKING. YOU CAN FEEL THE VIBRATIONS THAT CREATE SOUND.

CUP PHONE

CONNECT TWO PAPER CUPS WITH A STRING. SPEAK INTO ONE CUP AND LISTEN FROM THE OTHER. MAKE SURE THE STRING STAYS TIGHT.

ABOUT THE AUTHOR

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